

MD. MUHIUL ISLAM MUHIUDDIN

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ACADEMIC PROFILE

Electrical and Electronic Engineering graduate and current MSc student at BUET, simultaneously serving as Lecturer at BRAC University, Bangladesh. Research focuses on silicon photonics and photonic inverse design, developing adjoint topology-optimized silicon-on-insulator refractive index sensors using FDTD simulation and quantum sensing with VB^- spin defects in hexagonal boron nitride (h-BN). Author of 1 journal article in *IEEE Transactions on Electron Devices* and 2 IEEE conference papers. Brings teaching, curriculum development, and student mentoring experience, with strong competencies in nanophotonic and device simulation.

EDUCATION

Bangladesh University of Engineering and Technology (BUET)

Nov 2024 – Present

M.Sc. in Electrical and Electronic Engineering

GPA: **3.75 / 4.00** (Only defense remaining)

Thesis focus: Improving sensitivity of defect based hBN quantum sensor

Supervisor: Dr. Kawsar Alam, Professor, Dept. of EEE, BUET

Bangladesh University of Engineering and Technology (BUET)

Apr 2019 – Jul 2024

B.Sc. in Electrical and Electronic Engineering, Electronics Major

GPA: **3.93 / 4.00** Class Position: **12th** out of 199

RESEARCH EXPERIENCE

MSc Researcher, BUET

- Investigating ODMR-based quantum sensing with VB^- spin defects in hexagonal boron nitride (h-BN) for quantum magnetometry. Exploring adjoint-optimized photonic collection structures for enhanced spin readout efficiency.

Other Research work, BUET

- Designed and inversely optimized a compact all-silicon SOI refractive index sensor using adjoint-based topology optimization with FDTD simulation (Tidy3D). Achieved competitive sensitivity, quality factor, and figure of merit on a 220 nm SOI platform; manuscript in preparation for journal submission.

Undergraduate Researcher, BUET

Aug 2023 – Jul 2024

- Performed ab-initio DFT simulations (QuantumATK) of monolayer α -CS and SiP field-effect transistors. Investigated underlayer modulation strategies for performance scaling beyond 5-nm technology nodes; published in *IEEE Transactions on Electron Devices* (2025).

TEACHING EXPERIENCE

Lecturer, Dept. of Electrical and Electronic Engineering, BRAC University

Oct 2024 – Present

- Teach and coordinate Digital Electronics, Signals & Systems, and Energy Conversion Laboratory.
- Developed restructured lab manuals, prepared exam and assessment materials.
- Trainer** for a government-funded(SICIP) program (15 week long-ongoing) titled "**Analog VLSI Circuit and Layout Design**"

INDUSTRY EXPERIENCE

Trainee Engineer, Analog Design Team, Utkasemi Pvt. Limited, Bangladesh

Apr – Jul 2024

- Simulated and improved the behavior of existing analog circuit blocks using Cadence Virtuoso as part of a company-level circuit design project.

AWARDS AND RECOGNITION

- **University Merit Scholarship**, BUET 2019 – 2023
Awarded every semester for outstanding academic performance throughout the undergraduate programme.
- **Dean's List**, BUET 2019 – 2023
Recognised for consistently high academic standing across all terms.
- **1st Runners-Up**, VLSI Design Competition, EEE Day 2024, BUET 2024

PUBLICATIONS

Journal Article

- M.M.I. Muhiuddin, S. Kundu, and M.K. Alam, "Ultrascaled Monolayer α -CS and SiP 2-D n -Type FETs: Performance Enhancement via Underlayer Modulation Beyond 5-nm Technology Nodes," *IEEE Transactions on Electron Devices*, 2025.

Conference Papers

- S. Bain, S.B. Shuvo, S. Ahmed, S. Kundu, and M.M.I. Muhiuddin, "A High-Performance, Low-Noise Class-AB Folded-Cascode Op-Amp with Novel Biasing Circuit for Biomedical Applications," in *Proc. 27th Int. Conf. on Computer and Information Technology (ICCIT)*, 2024.
- M.M.I. Muhiuddin, A.A. Taki, K. Zahin, M.J. Karim, and S.M. AlKais, "Optimized Control Strategy for Self-Balancing Robot: Micro-Stepping Motors and PID Algorithm," in *Proc. IEEE 3rd Int. Conf. on Robotics, Automation, and Artificial Intelligence (RAAI)*, 2024.

SKILLS

Simulation & Photonics: Tidy3D (FDTD), QuantumATK, Quantum ESPRESSO, VASP, Cadence Virtuoso, NCSim, Material Studio, Lumerical (familiar), Simulink, EDA Playground

Programming: Python, MATLAB, C, C++, Verilog, SystemVerilog, ARM Assembly V7, HTML

Hardware Platforms: Altera FPGA, CPLD, STM32 Nucleo Development Board, ESP32 / Arduino, PCB Soldering

Languages: English , Bengali (native)

ACADEMIC PROJECTS

MSc and Undergraduate Projects

- Implemented a THz temporal differentiator reproducing a microring waveguide resonator paper; developed a baseband frequency-domain verification method using interpolated FDTD S21 data.
- Applied deep learning to building footprint segmentation (DMAF-RASNet architecture) and explored YOLOv5-based flood-affected people detection for remote-sensing applications.
- Low-Cost Real-Time Spectroscopy Using a Webcam and Python Analysis
- SPI Interface for RISC-V Using Verilog (FSM, RTL, Verification, Place & Route)
- Implementation of Single-Word Braille Display on FPGA
- Self-Balancing Robot: PID-Based Control System Implementation

LEADERSHIP AND PROFESSIONAL ENGAGEMENT

- **General Secretary**, Electron Devices Society, BUET 2023
Organised seminars, invited talks, and outreach events; managed student engagement activities.
- **Records Chair**, BUET Optical and Photonics Society 2022
Coordinated event documentation, outreach, and workshop activities.

REFERENCES

Dr. Kawsar Alam

Professor, Department of EEE

Bangladesh University of Engineering and Technology (BUET)

Dhaka-1000, Bangladesh